Single Port RAM Verification by using UVM

```
// Dut of single port ram
module dut_ram #(parameter DATA_WIDTH = 8,parameter ADDR_WIDTH = 4)(
        input  logic                clk,
        input  logic                rst_n,
      input  logic               write_en,
      input  logic [ADDR_WIDTH-1:0] addr,
      input  logic [DATA_WIDTH-1:0] data_in,
      output logic [DATA_WIDTH-1:0] data_out
);

logic [DATA_WIDTH-1:0] mem [0:(1<<ADDR_WIDTH)-1];

always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
                data_out <= '0;

                for (int i = 0; i < (1<<ADDR_WIDTH); i++) begin
                        mem[i] <= '0;
                end
        end
        else begin
                if (write_en) begin
                        mem[addr] <= data_in;
                end
                else begin
                        data_out <= mem[addr];
```

```
            end
        end
end


endmodule
```

Run-Command

[b23n_sarif_727@prac3 single_port_ram]$ ls

agent.sv dut_ram.fsdb novas.rc sequence_item.sv simv.daidir test.sv vdCovLog

coverage.sv environment.sv ram_if.sv sequencer.sv simv.vdb ucli.key

csrc monitor.sv ram_pkg.sv sequence.sv single_port_ram_design.sv vc_hdrs.h

driver.sv novas_dump.log scoreboard.sv simv tb_top.sv vdCov.conf

[b23n_sarif_727@prac3 single_port_ram]$ vcs -sverilog single_port_ram_design.sv ram_if.sv ram_pkg.sv tb_top.sv -ntb_opts uvm -kdb -lca -debug_access+all -full64

[b23n_sarif_727@prac3 single_port_ram]$ ./simv

[b23n_sarif_727@prac3 single_port_ram]$ ./simv -verdi

[b23n_sarif_727@prac3 single_port_ram]$ verdi -cov -covdir simv.vdb/

Testbench

```
// interface of single port ram

interface ram_if #(parameter DATA_WIDTH = 8, parameter ADDR_WIDTH = 4)(input logic clk);

    logic               rst_n;

   logic               write_en;

   logic [ADDR_WIDTH-1:0] addr;

   logic [DATA_WIDTH-1:0] data_in;

   logic [DATA_WIDTH-1:0] data_out;
```

```
        clocking drv_cb @(posedge clk);
                default input #1step output #1ns;
                output write_en, addr, data_in;
        endclocking

        clocking mon_cb @(posedge clk);
                default input #1step output #1ns;
                input write_en, addr, data_in, data_out;
        endclocking

        modport DRV (clocking drv_cb,output rst_n);
        modport MON (clocking mon_cb,input rst_n);

endinterface

// sequence_item of single port ram
class seq_item extends uvm_sequence_item;
        rand bit       write_en;
        rand bit [3:0] addr;
       rand bit [7:0] data_in;
        bit [7:0]      data_out;

        `uvm_object_utils_begin(seq_item)
        `uvm_field_int(write_en, UVM_ALL_ON)
        `uvm_field_int(addr,     UVM_ALL_ON)
        `uvm_field_int(data_in,  UVM_ALL_ON)
        `uvm_field_int(data_out, UVM_ALL_ON)
```

```
	`uvm_object_utils_end

	function new(string name = "seq_item");
		super.new(name);
	endfunction

endclass

//
class base_seq extends uvm_sequence #(seq_item);
	seq_item req;
	`uvm_object_utils(base_seq)

	function new(string name = "base_seq");
		super.new(name);
	endfunction

	task body();
		`uvm_info(get_type_name(), "Base sequence has been started:",
UVM_LOW)

		for (int i = 0; i < 16; i++) begin
			`uvm_do_with(req, { write_en == 1; addr == i; data_in == (8'hAA +
2*i);})
		end

		for (int i = 0; i < 16; i++) begin
			`uvm_do_with(req, { write_en == 0; addr == i;})
```

```
		end

		repeat(5) begin
			`uvm_do_with(req, { write_en== 1;})
		end

		repeat(5) begin
			`uvm_do_with(req, { write_en == 0;})
		end

		`uvm_info(get_type_name(), "All the Base sequence completely generated", UVM_LOW)
	endtask
endclass

//
class sequencer extends uvm_sequencer #(seq_item);
	`uvm_component_utils(sequencer)

	function new(string name = "sequencer", uvm_component parent = null);
		super.new(name, parent);
	endfunction

	function void build_phase(uvm_phase phase);
		super.build_phase(phase);
	endfunction

endclass
```

```
//
class driver extends uvm_driver #(seq_item);
	virtual ram_if.DRV vif;
	`uvm_component_utils(driver)

	function new(string name = "driver", uvm_component parent = null);
		super.new(name, parent);
	endfunction

	function void build_phase(uvm_phase phase);
		super.build_phase(phase);
		if (!uvm_config_db#(virtual ram_if.DRV)::get(this,"","vif",vif))
			`uvm_fatal(get_type_name(),"Virtual interface is not found from top level")
	endfunction

	task run_phase(uvm_phase phase);
		wait(vif.rst_n == 1'b1);
		@(vif.drv_cb);

		forever begin
			seq_item_port.get_next_item(req);
			@(vif.drv_cb);
			vif.drv_cb.write_en <= req.write_en;
		  vif.drv_cb.addr     <= req.addr;
		  vif.drv_cb.data_in  <= req.data_in;
```

```
                `uvm_info(get_type_name(),$sformatf("DRIVER: Write_en=%0d ADDR=%0d DATA_IN=%0h",req.write_en,req.addr,req.data_in), UVM_LOW)
                if(!req.write_en) begin
                        @(vif.drv_cb);
                end
                seq_item_port.item_done();
        end
    endtask

endclass

//
class monitor extends uvm_monitor;
    virtual ram_if.MON vif;
    uvm_analysis_port #(seq_item) mon_ap;
    `uvm_component_utils(monitor)

    function new(string name = "monitor",uvm_component parent = null);
        super.new(name, parent);
        mon_ap = new("mon_ap", this);
    endfunction

    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        if (!uvm_config_db#(virtual ram_if.MON)::get(this,"","vif",vif))
            `uvm_fatal(get_type_name(),"Virtual interface is not found from top level")
    endfunction
```

```
    task run_phase(uvm_phase phase);
        seq_item mon_item;
        forever begin
            @(vif.mon_cb);
            if (!vif.rst_n)
                continue;
            mon_item = seq_item::type_id::create("mon_item");

            mon_item.write_en = vif.mon_cb.write_en;
            mon_item.addr = vif.mon_cb.addr;
            mon_item.data_in = vif.mon_cb.data_in;

            if(!vif.mon_cb.write_en) begin
                @(vif.mon_cb);
            end
            mon_item.data_out = vif.mon_cb.data_out;
            `uvm_info(get_type_name(),$sformatf("MONITOR: Write_en=%0d ADDR=%0d DATA_IN=%0h DATA_OUT=%0h",mon_item.write_en,mon_item.addr,mon_item.data_in,mon_item.data_out), UVM_LOW)

            mon_ap.write(mon_item);
        end
    endtask

endclass
```

```
//
class scoreboard extends uvm_scoreboard;
	`uvm_component_utils(scoreboard)
	uvm_analysis_imp #(seq_item, scoreboard) sb_imp;

	bit [7:0] ref_mem [0:15];

	function new(string name = "scoreboard",uvm_component parent = null);
		super.new(name, parent);
		sb_imp = new("sb_imp", this);
	endfunction

	function void build_phase(uvm_phase phase);
		super.build_phase(phase);

		foreach (ref_mem[i]) begin
			ref_mem[i] = 8'h00;
		end
	endfunction

	function void write(seq_item req);

		if (req.write_en) begin
			ref_mem[req.addr] = req.data_in;
			`uvm_info("SCOREBOARD",$sformatf("WRITE PASS:
ADDR=%0d DATA=%0h",req.addr,req.data_in),UVM_LOW)
		end
		else begin
```

```
                    if (req.data_out === ref_mem[req.addr]) begin
                        `uvm_info("SCOREBOARD",$sformatf("READ PASS:
ADDR=%0d EXPECTED=%0h
ACTUAL=%0h",req.addr,ref_mem[req.addr],req.data_out),UVM_LOW)
                    end
                    else begin
                        `uvm_error("SCOREBOARD",$sformatf("READ FAIL:
ADDR=%0d EXPECTED=%0h
ACTUAL=%0h",req.addr,ref_mem[req.addr],req.data_out))
                    end
                end
        endfunction

endclass

//
class ram_coverage extends uvm_subscriber #(seq_item);
        `uvm_component_utils(ram_coverage)

        bit       cov_write_en;
      bit [3:0]  cov_addr;
      bit [7:0]  cov_data_in;

        covergroup ram_cg;

      cp_operation:
         coverpoint cov_write_en {
            bins read  = {0};
            bins write = {1};
```

```
    }

cp_address:
    coverpoint cov_addr {
        bins low_addr  = {[0:3]};
        bins mid_addr  = {[4:11]};
        bins high_addr = {[12:15]};
    }

cp_data_in:
    coverpoint cov_data_in {
        bins zero     = {8'h00};
        bins max      = {8'hFF};
        bins low_data = {[8'h01:8'h3F]};
        bins mid_data = {[8'h40:8'hBF]};
        bins high_data = {[8'hC0:8'hFE]};
    }

cross_operation_address:
    cross cp_operation, cp_address;

cross_operation_data:
    cross cp_operation, cp_data_in;

cross_write_addr_data:
    cross cp_operation, cp_address, cp_data_in;
```

```
    endgroup

    function new(string name = "ram_coverage",uvm_component parent = null);
            super.new(name, parent);
            ram_cg = new();
    endfunction

    function void write(seq_item t);
            cov_write_en = t.write_en;
          cov_addr     = t.addr;
          cov_data_in  = t.data_in;
            ram_cg.sample();
    endfunction

endclass
//
class agent extends uvm_agent;
        `uvm_component_utils(agent)

        driver    drv;
        sequencer seqr;
        monitor   mon;

        function new(string name = "agent", uvm_component parent = null);
                super.new(name,parent);
        endfunction
```

```
        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                drv = driver::type_id::create("drv", this);
                seqr =sequencer::type_id::create("seqr", this);
                mon =monitor::type_id::create("mon", this);
        endfunction

        function void connect_phase(uvm_phase phase);
                super.connect_phase(phase);
                drv.seq_item_port.connect(seqr.seq_item_export);
        endfunction

endclass

//
class env extends uvm_env;
        `uvm_component_utils(env)
        agent         agt;
        scoreboard    sb;
        ram_coverage  cov;

        function new(string name = "env", uvm_component parent = null);
                super.new(name, parent);
        endfunction

        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
```

```
        agt =agent::type_id::create("agt", this);
        sb =scoreboard::type_id::create("sb", this);
        cov =ram_coverage::type_id::create("cov", this);
    endfunction

    function void connect_phase(uvm_phase phase);
        super.connect_phase(phase);
        agt.mon.mon_ap.connect(sb.sb_imp);
        agt.mon.mon_ap.connect(cov.analysis_export);
    endfunction

endclass

//
class base_test extends uvm_test;
    `uvm_component_utils(base_test)
    env env_o;

    function new(string name = "base_test",uvm_component parent = null);
        super.new(name, parent);
    endfunction

    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        env_o =env::type_id::create("env_o",this);
    endfunction
```

```
	task run_phase(uvm_phase phase);
		base_seq bseq;
		phase.raise_objection(this);
		bseq =base_seq::type_id::create("bseq");
		bseq.start(env_o.agt.seqr);
		phase.drop_objection(this);
		`uvm_info(get_type_name(),"End of testcases",UVM_LOW)
	endtask

endclass

//
`include "uvm_macros.svh"
import uvm_pkg::*;
import ram_pkg::*;

module tb_top;

bit clk;
always #5 clk = ~clk;

ram_if #(.DATA_WIDTH(8),.ADDR_WIDTH(4)) vif(clk);
dut_ram #(.DATA_WIDTH(8),.ADDR_WIDTH(4)) DUT
(.clk(vif.clk),.rst_n(vif.rst_n),.write_en(vif.write_en),.addr(vif.addr),.data_in(vif.data_in),
.data_out(vif.data_out));

initial begin
	vif.rst_n = 1'b0;
```

```
        vif.write_en = 1'b0;
      vif.addr     = '0;
      vif.data_in  = '0;

        repeat(3) @(negedge clk);
        vif.rst_n = 1'b1;
end

initial begin
        uvm_config_db#(virtual ram_if.DRV)::set(null,"*","vif",vif);
        uvm_config_db#(virtual ram_if.MON)::set(null,"*","vif",vif);
end

initial begin
        run_test("base_test");
end

initial begin
        $fsdbDumpfile("dut_ram.fsdb");
      $fsdbDumpvars(0,tb_top);
end

endmodule

//
package ram_pkg;
```

```
  `include "uvm_macros.svh"
   import uvm_pkg::*;
  `include "sequence_item.sv"
`include "sequence.sv"
`include "sequencer.sv"
`include "driver.sv"
`include "monitor.sv"
`include "scoreboard.sv"
`include "coverage.sv"
`include "agent.sv"
`include "environment.sv"
`include "test.sv"

endpackage
```

Simulation Outputs

UVM_INFO sequence.sv(11) @ 0: uvm_test_top.env_o.agt.seqr@@bseq [base_seq] Base sequence has been started:

UVM_INFO monitor.sv(34) @ 45: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=0 DATA_IN=0 DATA_OUT=0

UVM_INFO scoreboard.sv(29) @ 45: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=0 EXPECTED=0 ACTUAL=0

UVM_INFO driver.sv(26) @ 45: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=0 DATA_IN=aa

UVM_INFO monitor.sv(34) @ 55: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=0 DATA_IN=aa DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 55: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=0 DATA=aa

UVM_INFO driver.sv(26) @ 55: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=1 DATA_IN=ac

UVM_INFO monitor.sv(34) @ 65: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=1 DATA_IN=ac DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 65: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=1 DATA=ac

UVM_INFO driver.sv(26) @ 65: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=2 DATA_IN=ae

UVM_INFO monitor.sv(34) @ 75: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=2 DATA_IN=ae DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 75: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=2 DATA=ae

UVM_INFO driver.sv(26) @ 75: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=3 DATA_IN=b0

UVM_INFO monitor.sv(34) @ 85: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=3 DATA_IN=b0 DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 85: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=3 DATA=b0

UVM_INFO driver.sv(26) @ 85: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=4 DATA_IN=b2

UVM_INFO monitor.sv(34) @ 95: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=4 DATA_IN=b2 DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 95: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=4 DATA=b2

UVM_INFO driver.sv(26) @ 95: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=5 DATA_IN=b4

UVM_INFO monitor.sv(34) @ 105: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=5 DATA_IN=b4 DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 105: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=5 DATA=b4

UVM_INFO driver.sv(26) @ 105: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=6 DATA_IN=b6

UVM_INFO monitor.sv(34) @ 115: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=6 DATA_IN=b6 DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 115: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=6 DATA=b6

UVM_INFO driver.sv(26) @ 115: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=7 DATA_IN=b8

UVM_INFO monitor.sv(34) @ 125: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=7 DATA_IN=b8 DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 125: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=7 DATA=b8

UVM_INFO driver.sv(26) @ 125: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=8 DATA_IN=ba

UVM_INFO monitor.sv(34) @ 135: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=8 DATA_IN=ba DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 135: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=8 DATA=ba

UVM_INFO driver.sv(26) @ 135: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=9 DATA_IN=bc

UVM_INFO monitor.sv(34) @ 145: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=9 DATA_IN=bc DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 145: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=9 DATA=bc

UVM_INFO driver.sv(26) @ 145: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=10 DATA_IN=be

UVM_INFO monitor.sv(34) @ 155: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=10 DATA_IN=be DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 155: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=10 DATA=be

UVM_INFO driver.sv(26) @ 155: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=11 DATA_IN=c0

UVM_INFO monitor.sv(34) @ 165: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=11 DATA_IN=c0 DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 165: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=11 DATA=c0

UVM_INFO driver.sv(26) @ 165: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=12 DATA_IN=c2

UVM_INFO monitor.sv(34) @ 175: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=12 DATA_IN=c2 DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 175: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=12 DATA=c2

UVM_INFO driver.sv(26) @ 175: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=13 DATA_IN=c4

UVM_INFO monitor.sv(34) @ 185: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=13 DATA_IN=c4 DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 185: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=13 DATA=c4

UVM_INFO driver.sv(26) @ 185: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=14 DATA_IN=c6

UVM_INFO monitor.sv(34) @ 195: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=14 DATA_IN=c6 DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 195: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=14 DATA=c6

UVM_INFO driver.sv(26) @ 195: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=15 DATA_IN=c8

UVM_INFO monitor.sv(34) @ 205: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=15 DATA_IN=c8 DATA_OUT=0

UVM_INFO scoreboard.sv(25) @ 205: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=15 DATA=c8

UVM_INFO driver.sv(26) @ 205: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=0 DATA_IN=b4

UVM_INFO monitor.sv(34) @ 225: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=0 DATA_IN=b4 DATA_OUT=aa

UVM_INFO scoreboard.sv(29) @ 225: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=0 EXPECTED=aa ACTUAL=aa

UVM_INFO driver.sv(26) @ 225: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=1 DATA_IN=e0

UVM_INFO monitor.sv(34) @ 245: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=1 DATA_IN=e0 DATA_OUT=ac

UVM_INFO scoreboard.sv(29) @ 245: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=1 EXPECTED=ac ACTUAL=ac

UVM_INFO driver.sv(26) @ 245: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=2 DATA_IN=c2

UVM_INFO monitor.sv(34) @ 265: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=2 DATA_IN=c2 DATA_OUT=ae

UVM_INFO scoreboard.sv(29) @ 265: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=2 EXPECTED=ae ACTUAL=ae

UVM_INFO driver.sv(26) @ 265: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=3 DATA_IN=69

UVM_INFO monitor.sv(34) @ 285: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=3 DATA_IN=69 DATA_OUT=b0

UVM_INFO scoreboard.sv(29) @ 285: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=3 EXPECTED=b0 ACTUAL=b0

UVM_INFO driver.sv(26) @ 285: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=4 DATA_IN=1b

UVM_INFO monitor.sv(34) @ 305: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=4 DATA_IN=1b DATA_OUT=b2

UVM_INFO scoreboard.sv(29) @ 305: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=4 EXPECTED=b2 ACTUAL=b2

UVM_INFO driver.sv(26) @ 305: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=5 DATA_IN=27

UVM_INFO monitor.sv(34) @ 325: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=5 DATA_IN=27 DATA_OUT=b4

UVM_INFO scoreboard.sv(29) @ 325: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=5 EXPECTED=b4 ACTUAL=b4

UVM_INFO driver.sv(26) @ 325: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=6 DATA_IN=f0

UVM_INFO monitor.sv(34) @ 345: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=6 DATA_IN=f0 DATA_OUT=b6

UVM_INFO scoreboard.sv(29) @ 345: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=6 EXPECTED=b6 ACTUAL=b6

UVM_INFO driver.sv(26) @ 345: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=7 DATA_IN=97

UVM_INFO monitor.sv(34) @ 365: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=7 DATA_IN=97 DATA_OUT=b8

UVM_INFO scoreboard.sv(29) @ 365: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=7 EXPECTED=b8 ACTUAL=b8

UVM_INFO driver.sv(26) @ 365: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=8 DATA_IN=55

UVM_INFO monitor.sv(34) @ 385: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=8 DATA_IN=55 DATA_OUT=ba

UVM_INFO scoreboard.sv(29) @ 385: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=8 EXPECTED=ba ACTUAL=ba

UVM_INFO driver.sv(26) @ 385: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=9 DATA_IN=ef

UVM_INFO monitor.sv(34) @ 405: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=9 DATA_IN=ef DATA_OUT=bc

UVM_INFO scoreboard.sv(29) @ 405: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=9 EXPECTED=bc ACTUAL=bc

UVM_INFO driver.sv(26) @ 405: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=10 DATA_IN=c1

UVM_INFO monitor.sv(34) @ 425: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=10 DATA_IN=c1 DATA_OUT=be

UVM_INFO scoreboard.sv(29) @ 425: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=10 EXPECTED=be ACTUAL=be

UVM_INFO driver.sv(26) @ 425: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=11 DATA_IN=96

UVM_INFO monitor.sv(34) @ 445: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=11 DATA_IN=96 DATA_OUT=c0

UVM_INFO scoreboard.sv(29) @ 445: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=11 EXPECTED=c0 ACTUAL=c0

UVM_INFO driver.sv(26) @ 445: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=12 DATA_IN=fd

UVM_INFO monitor.sv(34) @ 465: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=12 DATA_IN=fd DATA_OUT=c2

UVM_INFO scoreboard.sv(29) @ 465: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=12 EXPECTED=c2 ACTUAL=c2

UVM_INFO driver.sv(26) @ 465: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=13 DATA_IN=7f

UVM_INFO monitor.sv(34) @ 485: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=13 DATA_IN=7f DATA_OUT=c4

UVM_INFO scoreboard.sv(29) @ 485: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=13 EXPECTED=c4 ACTUAL=c4

UVM_INFO driver.sv(26) @ 485: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=14 DATA_IN=e7

UVM_INFO monitor.sv(34) @ 505: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=14 DATA_IN=e7 DATA_OUT=c6

UVM_INFO scoreboard.sv(29) @ 505: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=14 EXPECTED=c6 ACTUAL=c6

UVM_INFO driver.sv(26) @ 505: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=15 DATA_IN=53

UVM_INFO monitor.sv(34) @ 525: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=15 DATA_IN=53 DATA_OUT=c8

UVM_INFO scoreboard.sv(29) @ 525: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=15 EXPECTED=c8 ACTUAL=c8

UVM_INFO driver.sv(26) @ 525: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=1 DATA_IN=f4

UVM_INFO monitor.sv(34) @ 535: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=1 DATA_IN=f4 DATA_OUT=c8

UVM_INFO scoreboard.sv(25) @ 535: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=1 DATA=f4

UVM_INFO driver.sv(26) @ 535: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=7 DATA_IN=40

UVM_INFO monitor.sv(34) @ 545: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=7 DATA_IN=40 DATA_OUT=c8

UVM_INFO scoreboard.sv(25) @ 545: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=7 DATA=40

UVM_INFO driver.sv(26) @ 545: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=3 DATA_IN=7c

UVM_INFO monitor.sv(34) @ 555: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=3 DATA_IN=7c DATA_OUT=c8

UVM_INFO scoreboard.sv(25) @ 555: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=3 DATA=7c

UVM_INFO driver.sv(26) @ 555: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=11 DATA_IN=d3

UVM_INFO monitor.sv(34) @ 565: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=11 DATA_IN=d3 DATA_OUT=c8

UVM_INFO scoreboard.sv(25) @ 565: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=11 DATA=d3

UVM_INFO driver.sv(26) @ 565: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=1 ADDR=12 DATA_IN=73

UVM_INFO monitor.sv(34) @ 575: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=1 ADDR=12 DATA_IN=73 DATA_OUT=c8

UVM_INFO scoreboard.sv(25) @ 575: uvm_test_top.env_o.sb [SCOREBOARD] WRITE PASS: ADDR=12 DATA=73

UVM_INFO driver.sv(26) @ 575: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=10 DATA_IN=72

UVM_INFO monitor.sv(34) @ 595: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=10 DATA_IN=72 DATA_OUT=be

UVM_INFO scoreboard.sv(29) @ 595: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=10 EXPECTED=be ACTUAL=be

UVM_INFO driver.sv(26) @ 595: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=4 DATA_IN=0

UVM_INFO monitor.sv(34) @ 615: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=4 DATA_IN=0 DATA_OUT=b2

UVM_INFO scoreboard.sv(29) @ 615: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=4 EXPECTED=b2 ACTUAL=b2

UVM_INFO driver.sv(26) @ 615: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=7 DATA_IN=ce

UVM_INFO monitor.sv(34) @ 635: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=7 DATA_IN=ce DATA_OUT=40

UVM_INFO scoreboard.sv(29) @ 635: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=7 EXPECTED=40 ACTUAL=40

UVM_INFO driver.sv(26) @ 635: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=15 DATA_IN=b0

UVM_INFO monitor.sv(34) @ 655: uvm_test_top.env_o.agt.mon [monitor] MONITOR: Write_en=0 ADDR=15 DATA_IN=b0 DATA_OUT=c8

UVM_INFO scoreboard.sv(29) @ 655: uvm_test_top.env_o.sb [SCOREBOARD] READ PASS: ADDR=15 EXPECTED=c8 ACTUAL=c8

UVM_INFO driver.sv(26) @ 655: uvm_test_top.env_o.agt.drv [driver] DRIVER: Write_en=0 ADDR=8 DATA_IN=27

UVM_INFO sequence.sv(29) @ 665: uvm_test_top.env_o.agt.seqr@@bseq [base_seq] All the Base sequence completely generated

UVM_INFO test.sv(21) @ 665: uvm_test_top [base_test] End of testcases

UVM_INFO /usr/synopsys/vcs/W-2024.09-1/etc/uvm/base/uvm_objection.svh(1274) @ 665: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase

--- UVM Report Summary ---

** Report counts by severity

UVM_INFO : 131

UVM_WARNING : 0

UVM_ERROR : 0

UVM_FATAL : 0

** Report counts by id

[RNTST] 1

[SCOREBOARD] 42

[TEST_DONE] 1

[base_seq] 2

[base_test] 1

[driver] 42

[monitor] 42

$finish called from file "/usr/synopsys/vcs/W-2024.09-1/etc/uvm/base/uvm_root.svh", line 437.

$finish at simulation time 665

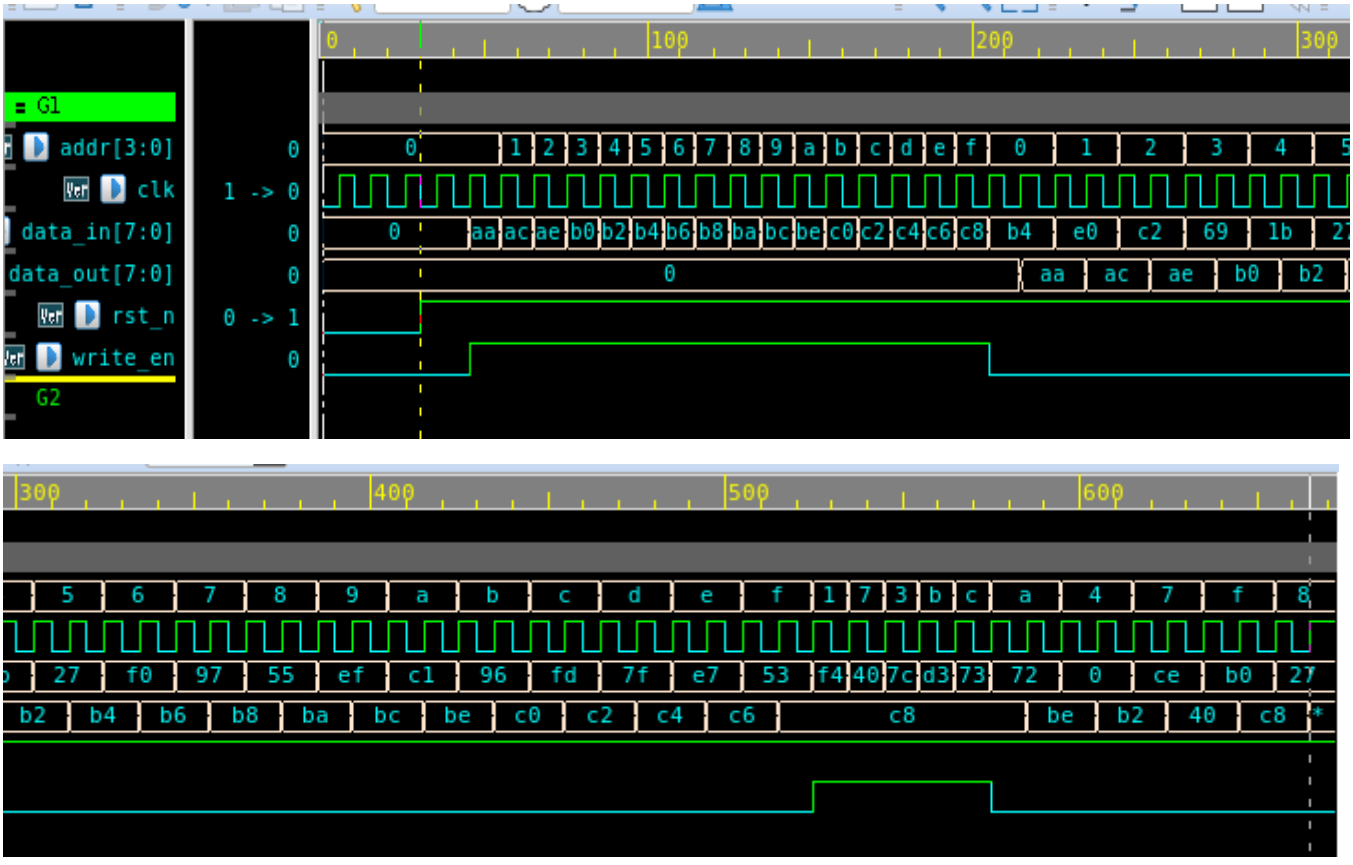
V C S S i m u l a t i o n R e p o r t

Time: 665

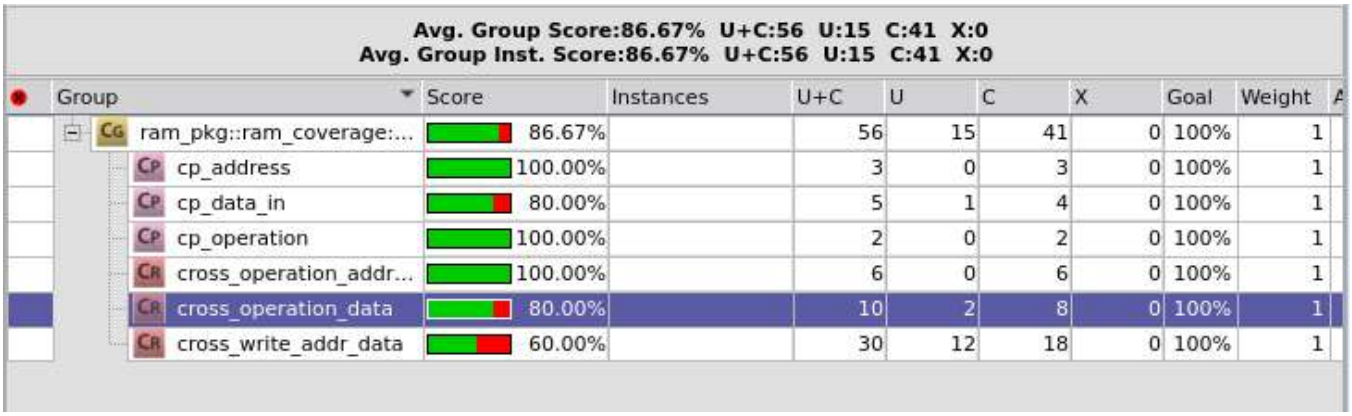
CPU Time: 0.170 seconds; Data structure size: 0.6Mb

Wed Aug 28 15:50:04 2026

Simulation waveform

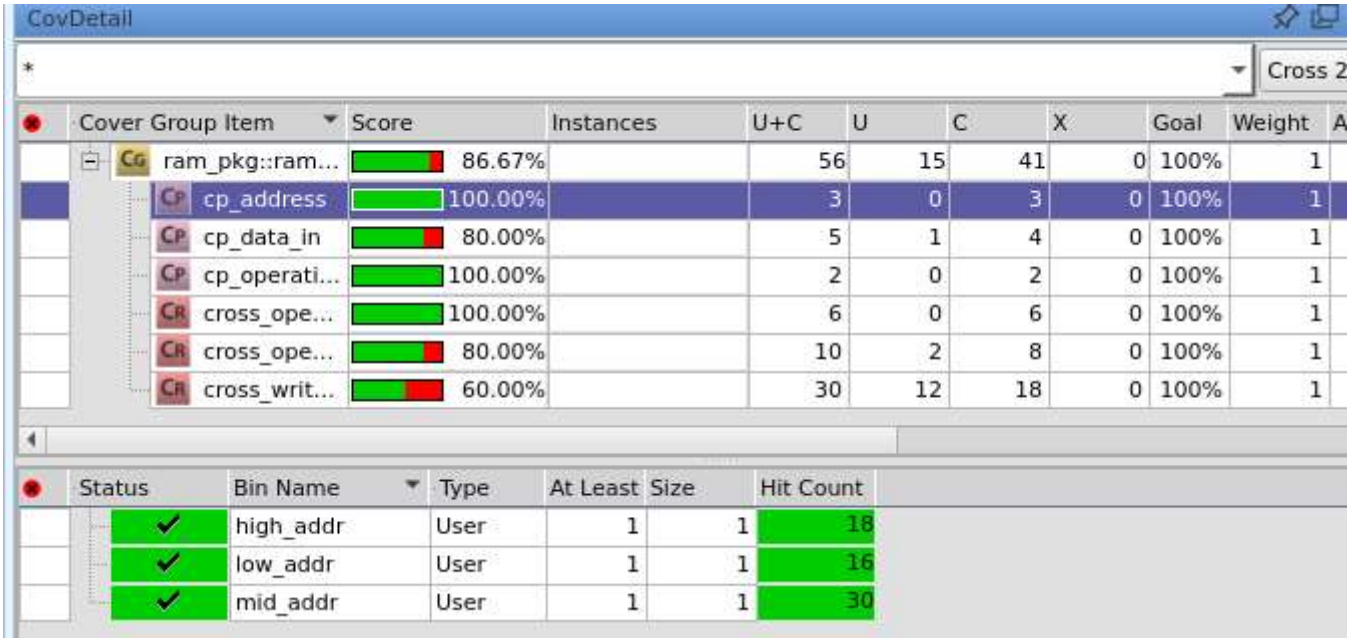


Coverage Report:-



Avg. Group Score:86.67% U+C:56 U:15 C:41 X:0
Avg. Group Inst. Score:86.67% U+C:56 U:15 C:41 X:0

Group	Score	Instances	U+C	U	C	X	Goal	Weight
ram_pkg::ram_coverage:...	86.67%		56	15	41	0	100%	1
cp_address	100.00%		3	0	3	0	100%	1
cp_data_in	80.00%		5	1	4	0	100%	1
cp_operation	100.00%		2	0	2	0	100%	1
cross_operation_addr...	100.00%		6	0	6	0	100%	1
cross_operation_data	80.00%		10	2	8	0	100%	1
cross_write_addr_data	60.00%		30	12	18	0	100%	1



CovDetail

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Cross 2

Cover Group Item	Score	Instances	U+C	U	C	X	Goal	Weight
ram_pkg::ram...	86.67%		56	15	41	0	100%	1
cp_address	100.00%		3	0	3	0	100%	1
cp_data_in	80.00%		5	1	4	0	100%	1
cp_operati...	100.00%		2	0	2	0	100%	1
cross_ope...	100.00%		6	0	6	0	100%	1
cross_ope...	80.00%		10	2	8	0	100%	1
cross_writ...	60.00%		30	12	18	0	100%	1

Status	Bin Name	Type	At Least	Size	Hit Count
✓	high_addr	User	1	1	18
✓	low_addr	User	1	1	16
✓	mid_addr	User	1	1	30